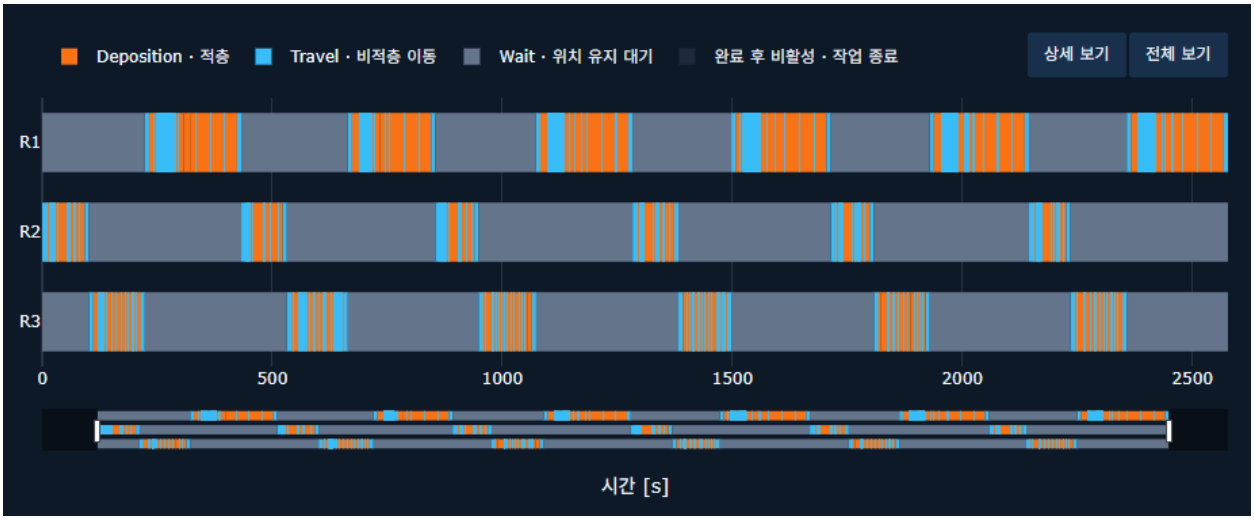
d 름		**학번**	**2022089761**
깃허브 링크			

- 알고리즘 1 번 | 1.0.2 Main Layer PPO
  - 모델별 적용 결과

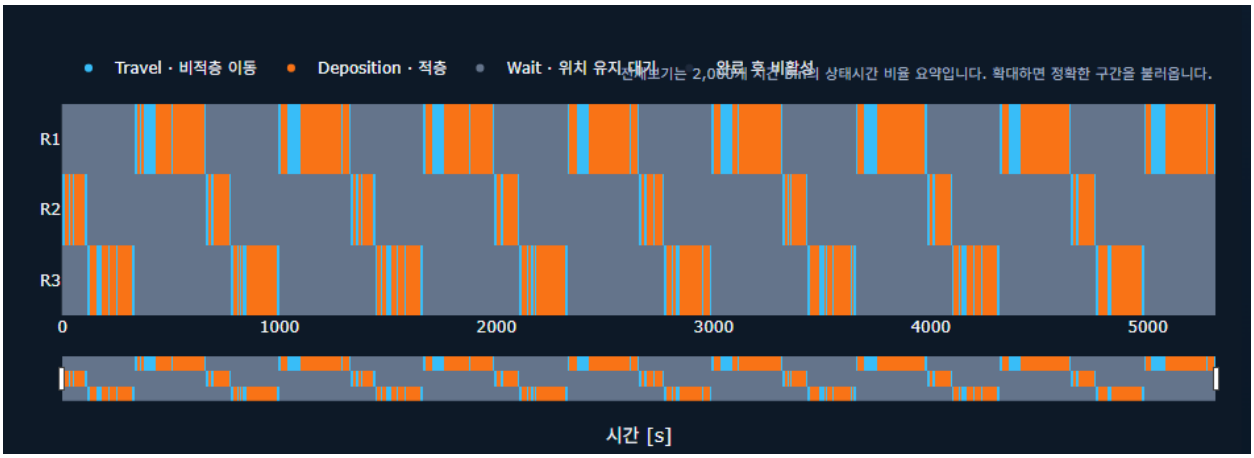
모델명	**Makespan (DD:HH:MM:SS)**	**계산 시간 (DD:HH:MM:SS)**	**Validator 적용 결과 (Pass or Fail)**
01 (Honeycomb)	00:00:42:57	00:00:00:07	PASS
02 (Spiral)	00:01:28:31	00:00:00:12	PASS
03 (Triangular truss)	00:01:07:50	00:00:00:05	PASS
04 (Motor inner)	00:20:06:15	00:00:01:07	PASS
05 (Twisted shell)	05:01:23:21	00:00:08:35	PASS
06 (Angle frame)	00:02:40:19	00:00:00:23	PASS

  - Validator 적용 결과 캡쳐 화면

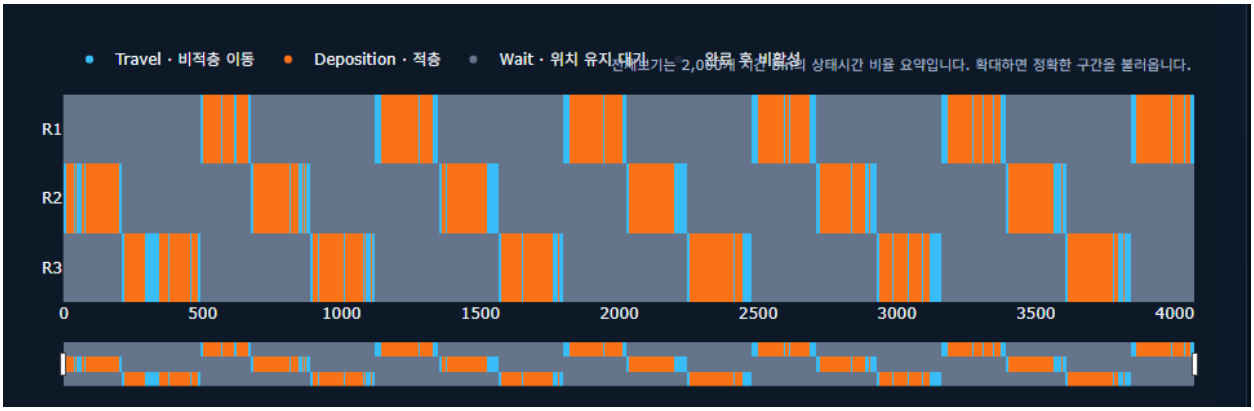
◆ 01



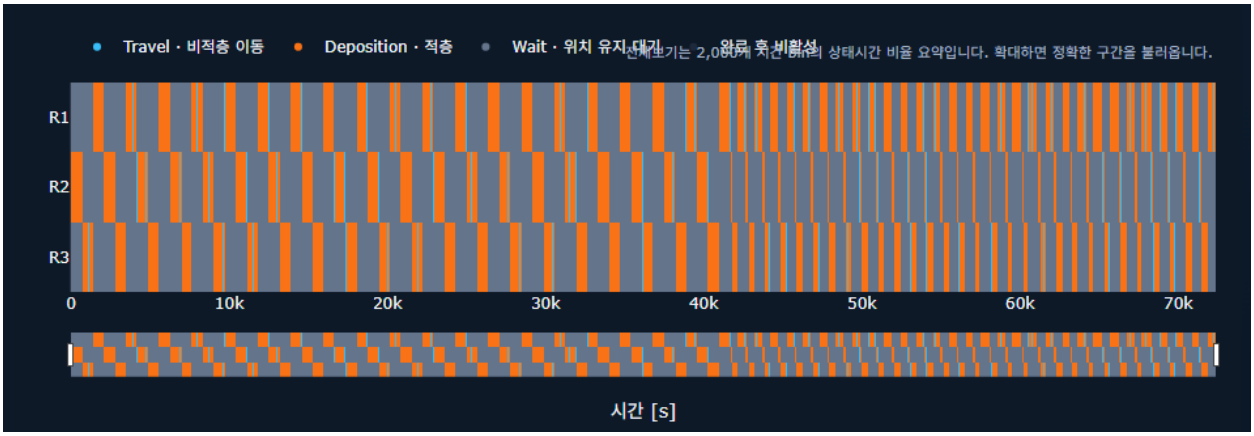
02


Travel · 비적층 이동
Deposition · 적층
Wait · 위치 유지 대기
R1
R2
R3
0
1000
2000
3000
4000
5000
시간 [s]

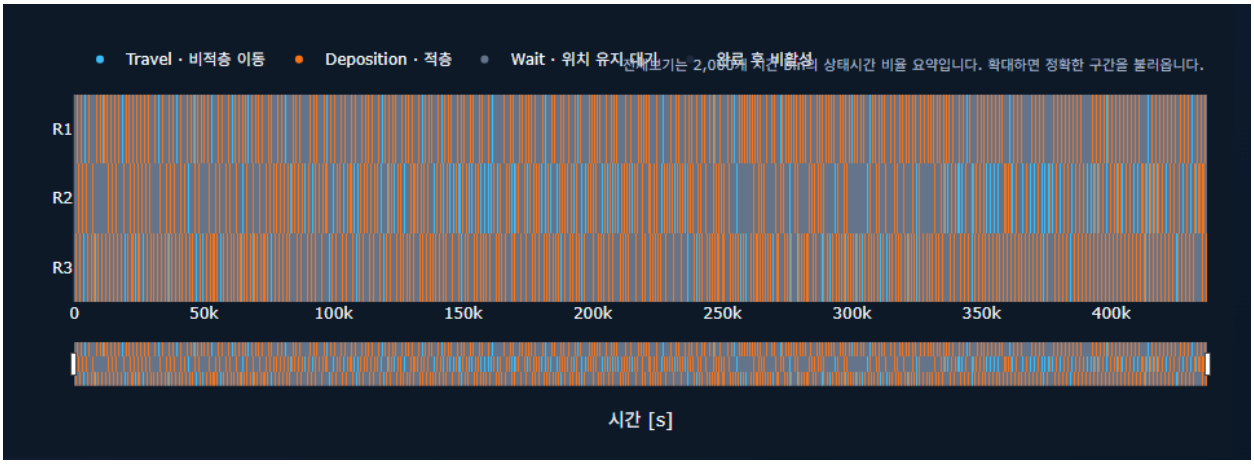
03


Travel · 비적층 이동
Deposition · 적층
Wait · 위치 유지 대기
R1
R2
R3
0
500
1000
1500
2000
2500
3000
3500
4000

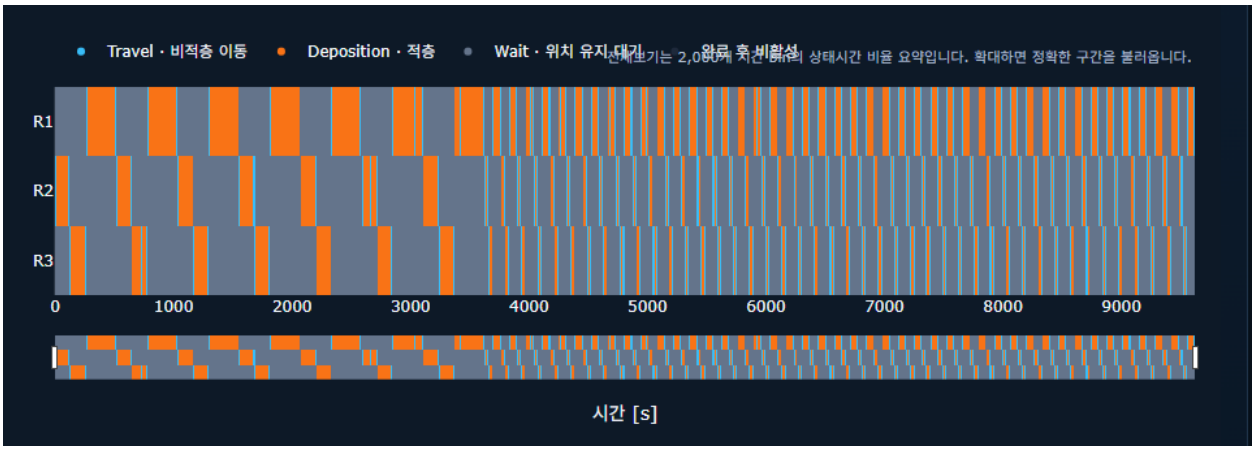
◆ 04


Travel · 비적층 이동
Deposition · 적층
Wait · 위치 유지 대기
R1
R2
R3
0
10k
20k
30k
40k
50k
60k
70k
시간 [s]

◆ 05


Travel · 비적층 이동
Deposition · 적층
Wait · 위치 유지 대기
전체보기는 2,000개 처리 후의 상태시간 비율 요약입니다. 확대하면 정확한 구간을 불러옵니다.
R1
R2
R3
0
50k
100k
150k
200k
250k
300k
350k
400k
시간 [s]

● 06


Travel · 비적층 이동
Deposition · 적층
Wait · 위치 유지 대기
전체보기는 2,000개 처리 후의 상태시간 비율 요약입니다. 확대하면 정확한 구간을 불러옵니다.
R1
R2
R3
0
1000
2000
3000
4000
5000
6000
7000
8000
9000
시간 [s]

수도코드

Main 1.0.2 - training and environment

Paper-style algorithm | STL preprocessing, layer observations, PPO actions, idle-time reward, and final Validator

Algorithm 1. Main 1.0.2 - preprocessing and PPO training

```
Input: training STL/config set D_tr; validation set D_val; rollout length n_steps
For each scenario q in D_tr and D_val, parse config.yaml and validate robot/process parameters
Load target.stl with Trimesh; concatenate scene meshes; merge vertices and duplicate faces
Remove degenerate faces and unused vertices; repair normals; optionally fill holes
Check finite coordinates, watertight policy, build-plane bounds, and mesh validity
Choose evaluation layer indices; slice at build_plane + (k + 0.5) * layer_height
Normalize each section polygon using snap tolerance and area epsilon; discard empty layers
Cache a 16 x 16 fractional-area raster for each target layer
Initialize 3 robots at home; initialize PPO policy pi_theta, MultiDiscrete([8,3,4,2,2,3,6])
For PPO update u = 1, ..., U do
    Reset a training STL episode; initialize schedule and cumulative geometry state
    For each layer k do
        Observe progress, makespan, IoU, layer bounds/area/perimeter/raster and robot/process limits
        Sample action: angle, spacing, corner, reversal, partition axis/quota, robot order
        Generate inset contours and clipped hatch; split long segments; order from chosen corner
        Assign segments to reachable robots; rebalance unreachable requested assignments
        Build travel/deposition/wait routes; validate reach, trajectory rules and collisions
        On collision retry the same layer with serialized starts; reject if still infeasible
        Compute this layer deposited geometry, layer IoU and duration
        Compute robot activity utilization u_r and idle_factor = 1 + sum(idle_r)/duration
        r_layer = IoU - 0.1 * duration * idle_factor / reference_time
        On final layer run full WAAM Validator; add +1 for PASS or -1 for FAIL/ERROR
        Append layer routes with time offset; update observation and PPO rollout buffer
    end for; update theta with PPO; periodically score validation checkpoints by pass rate
end for; export best policy, trajectories, validation JSON and Gantt charts
```

Procedure 1. Environment preprocessing and one-layer transition

```
ENV_INIT(q): parse config; clean and validate STL mesh; determine and slice requested layers
    Store normalized polygons, layer heights, robot bases/homes, reach and collision limits
    Build observation schema; lazily cache each layer 16 x 16 raster
LAYER_STEP(state, action): decode 7 action fields and select current target layer
    Generate path; assign segments; construct synchronized T/D/W routes for 3 robots
    Check local reach and collision; retry serialized schedule or reject with invalid reward -2
    Compute local deposited geometry and shape metrics; update cumulative geometry and IoU
    Compute reward = IoU - 0.1 * duration * (1 + sum(idle)/duration) / reference_time
    If final layer, export STL/config/trajectory bundle and call official WAAM Validator
    Add terminal reward (+1 PASS, -1 FAIL/ERROR); return next observation, reward and done
```

Notation and implementation notes

- Preprocessing follows load_target_mesh, determine_evaluation_layers, slice_target_layers, and normalize_polygon.
- Observation includes layer geometry and a cached raster plus schedule, robot, process, and validation features.
- One MultiDiscrete action (7 fields) generates one complete layer path.
- Idle factor is 3 when one robot is active for the full layer, it is 1 when all three are active.
- Local checks reject infeasible candidates; the official whole-trajectory Validator is final authority.

- ㄴ이 알고리즘을 이 문제에 적용한 이유
  - ◆
- 알고리즘에 대한 세부 설명(특징, 장단점 등)
  - ◆

■ 기타

■ Parameters

파라미터	현재 설정	의미
학습 업데이트	- / -회	요청 횟수 / 완료 횟수
학습 타임스텝	1024 / 1024	완료 / 요청
n_steps	64	PPO rollout 단계 수
batch_size	32	미니배치 크기
learning_rate	0.0003	학습률
gamma	0.99	할인율
gae_lambda	0.95	GAE 람다
seed	42	재현용 시드
device	cuda	정책 네트워크 계산 장치
early_stopping	-	조기 종료 설정
eval_every_updates	4	평가 주기
eval_episodes	1	평가 STL 별 반복 횟수
geometry_device	cpu	기하·Validator 계산 장치
training STL	6	학습 입력 STL 개수
validation STL	0	학습 중 평가 STL 개수
placement_enabled	False	전역 XY placement PPO 사용 여부
reward_formula	-	학습 보상식 ID
best_step	256	최고 검증 점수 체크포인트 단계
best_model	best_model.zip	실행 시 불러오는 모델
training_status	completed	학습 기록 상태

■

■

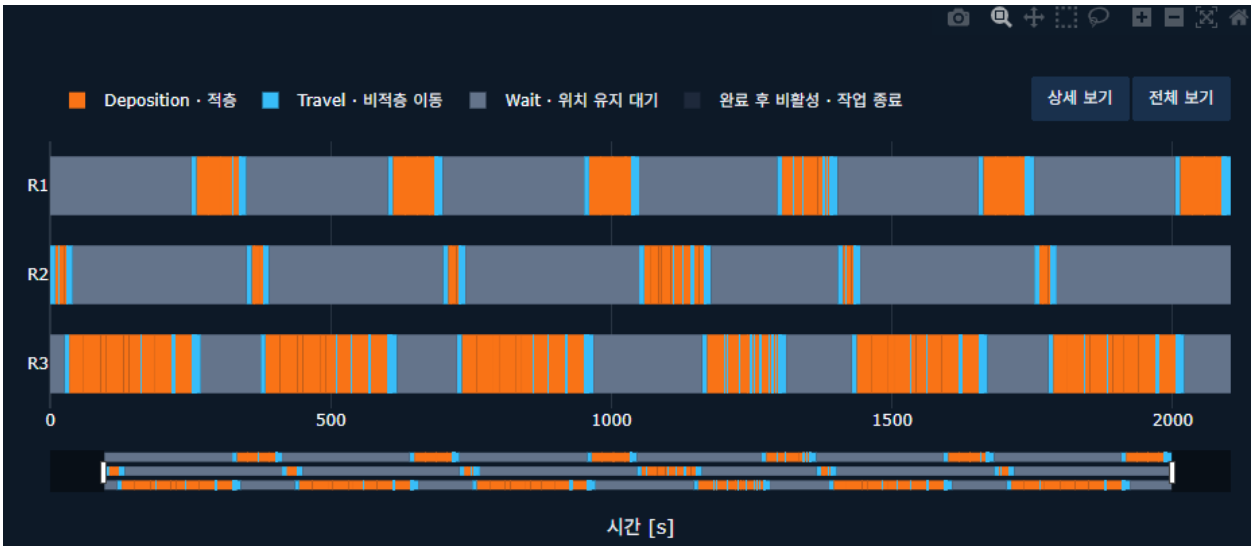
- **알고리즘 2 번**
  - ■

- 알고리즘 1 번 | Minor1 Placement and Scheduler PPO
  - 모델별 적용 결과

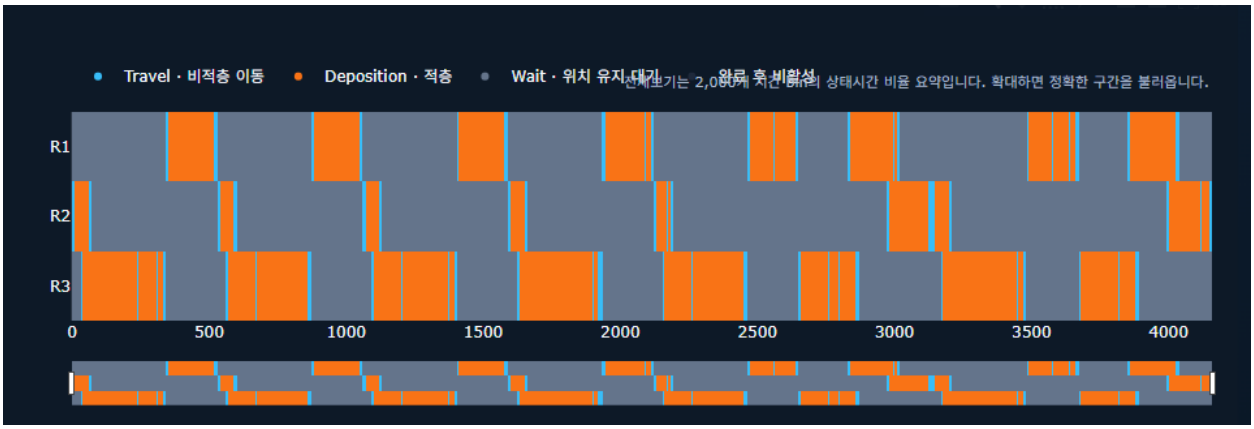
모델명	Makespan (DD:HH:MM:SS)	계산 시간 (DD:HH:MM:SS)	Validator 적용 결과 (Pass or Fail)
01 (Honeycomb)	00:00:35:03	00:00:00:13	PASS
02 (Spiral)	00:01:09:18	00:00:00:25	PASS
03 (Triangular truss)	00:00:57:30	00:00:00:14	PASS
04 (Motor inner)	00:15:02:28	00:00:02:59	PASS
05 (Twisted shell)	02:03:13:53	00:00:37:06	PASS
06 (Angle frame)	00:02:12:27	00:00:01:26	PASS

  - Validator 적용 결과 캡쳐 화면

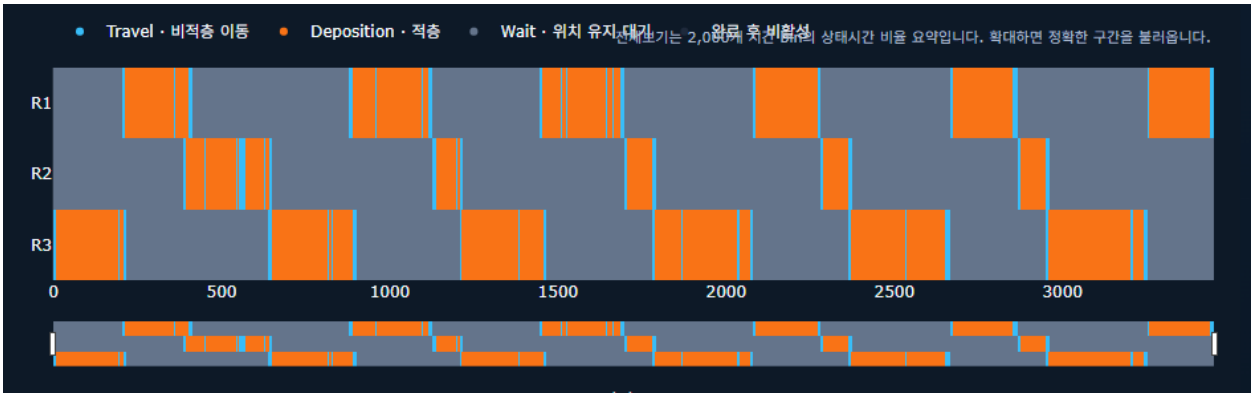
◆ 01



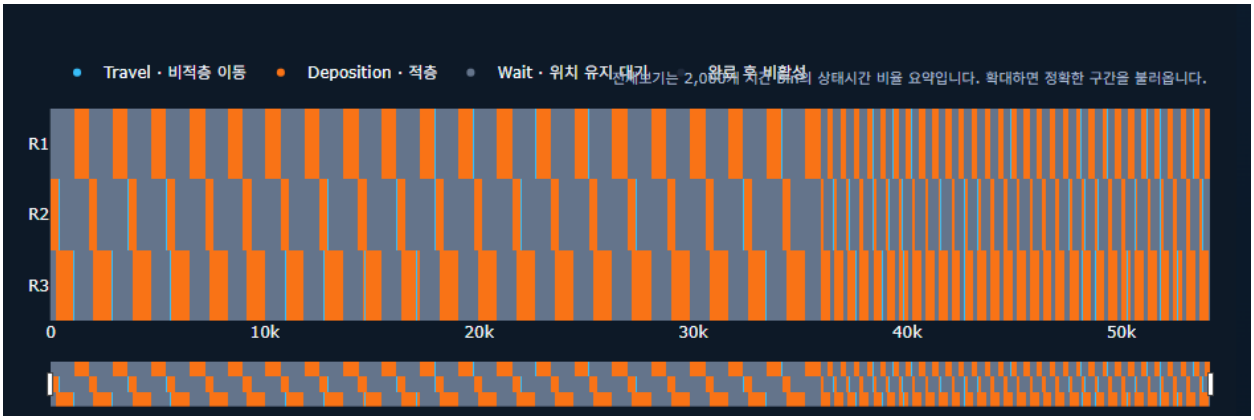
◆ 02



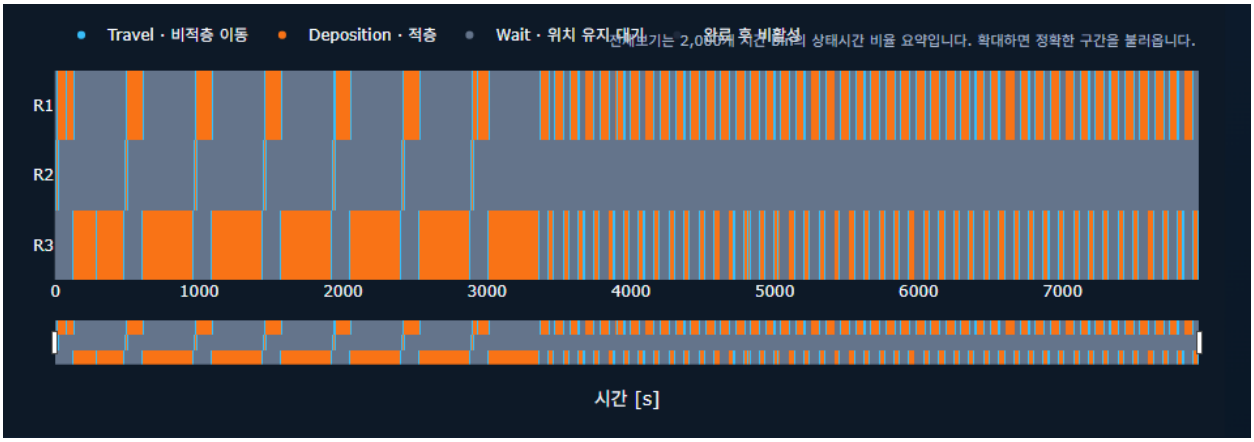
◆ 03


Travel · 비적층 이동
Deposition · 적층
Wait · 위치 유지 대기
완료 후 비활성
선택 구간의 상태시간 비율 요약입니다. 확대하면 정확한 구간을 불러옵니다.
R1
R2
R3
0
500
1000
1500
2000
2500
3000

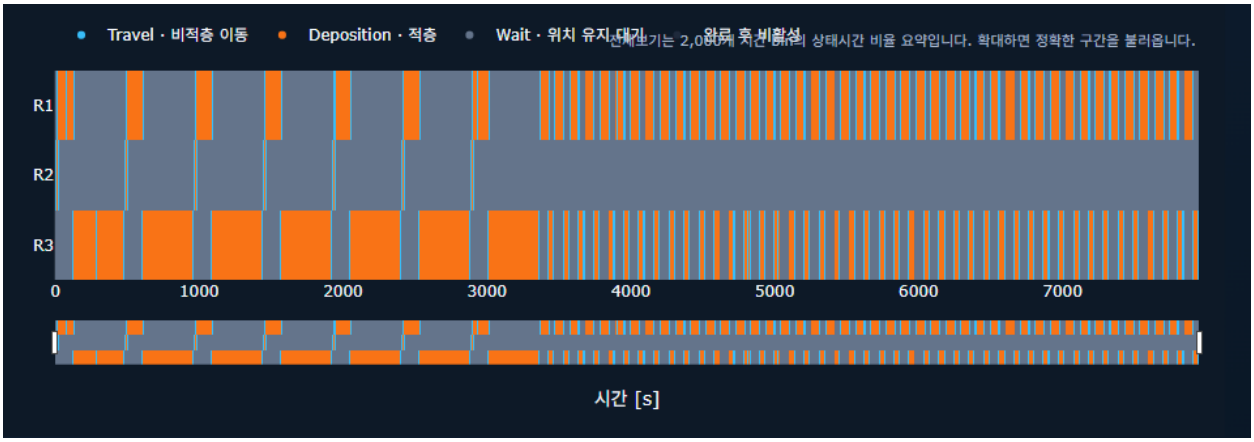
◆ 04


Travel · 비적층 이동
Deposition · 적층
Wait · 위치 유지 대기
완료 후 비활성
선택 구간의 상태시간 비율 요약입니다. 확대하면 정확한 구간을 불러옵니다.
R1
R2
R3
0
10k
20k
30k
40k
50k

◆ 05


Travel · 비적층 이동
Deposition · 적층
Wait · 위치 유지 대기
완료 후 비활성
선택 구간의 상태시간 비율 요약입니다. 확대하면 정확한 구간을 불러옵니다.
R1
R2
R3
0
1000
2000
3000
4000
5000
6000
7000
시간 [s]

◆ 06


Travel · 비적층 이동
Deposition · 적층
Wait · 위치 유지 대기
R1
R2
R3
0
1000
2000
3000
4000
5000
6000
7000
시간 [s]

● 수도코드

Algorithm 1. Minor 1 — joint path and placement PPO

```
Input: training STLs D_tr, validation set D_val, updates K
Initialize path policy πθ from compatible 1.0.2 weights
Initialize placement policy πϕ; placement action is global (Δx, Δy)
for update k = 1, …, K do
    Freeze πϕ; collect 64 layer steps using πθ and fixed placement
    Update θ with PPO using the layer rewards R_layer
    Freeze πθ
    for b = 1, …, 2 STL episodes do
        Select one STL q and predict Δxy ← πϕ(observe(q))
        Translate all of q by Δxy before slicing; preserve Z and shape
        Execute πθ for every layer; collect {R_layer} for STL q
        Set R_place ← mean({R_layer} from this STL episode only)
    end for; update ϕ with PPO using R_place
    if k mod 4 = 0 then evaluate (πθ, πϕ) on D_val
        Select best pair: pass rate, normalized makespan, overlap
    end if
end for
return best path-policy / placement-policy pair
```

- ㄴ이 알고리즘을 이 문제에 적용한 이유
  - ◆
- 알고리즘에 대한 세부 설명(특징, 장단점 등)
  - ◆

■ 기타

■ Parameters

파라미터	현재 설정	의미
학습 업데이트	28 / 40 회	요청 횟수 / 완료 횟수
학습 타임스텝	1792 / 2560	완료 / 요청
n_steps	64	PPO rollout 단계 수
batch_size	32	미니배치 크기
learning_rate	0.0003	학습률
gamma	0.99	할인율
gae_lambda	0.95	GAE 람다
seed	42	재현용 시드
device	cuda	정책 네트워크 계산 장치
early_stopping	False	조기 종료 설정
eval_every_updates	4	평가 주기
eval_episodes	1	평가 STL 별 반복 횟수
geometry_device	cpu	기하·Validator 계산 장치
training STL	6	학습 입력 STL 개수
validation STL	6	학습 중 평가 STL 개수

placement_enabled	True	전역 XY placement PPO 사용
placement_steps_per_update	2	업데이트당 placement episode 수
placement_every	1	placement 학습 주기
placement PPO n_steps	2	placement rollout 길이
placement PPO n_epochs	4	placement PPO 반복 횟수
reward_formula	iou_gt_0.9_utilization_over_makespan_v1	layer 보상식 ID
placement_reward	mean_of_layer_rewards_from_this_stl_only	동일 STL 의 layer reward 평균
best_step	1024	최고 검증 점수 체크포인트 단계
best_model	best_model.zip	실행 시 불러오는 모델
training_status	started	학습 기록 상태



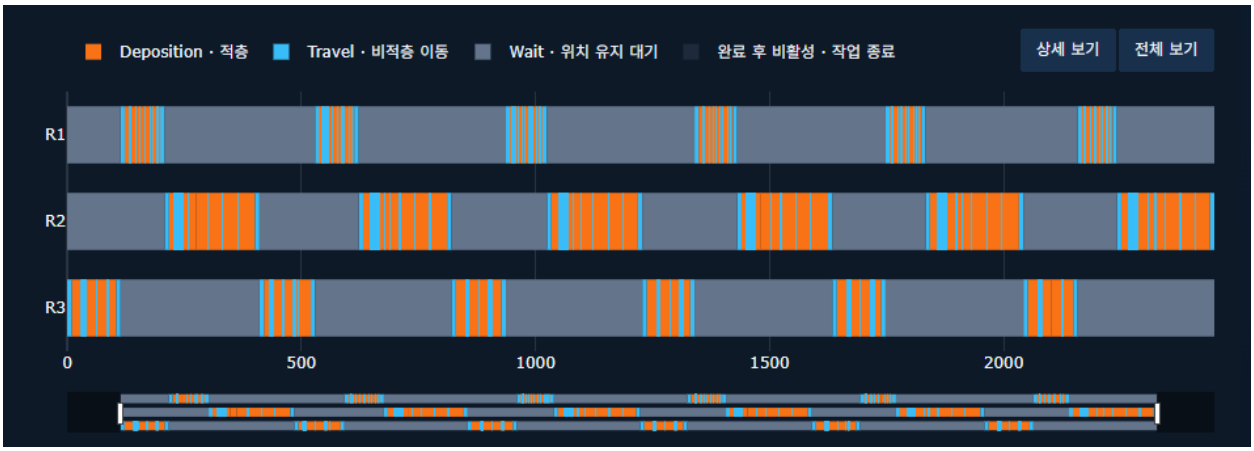
- **알고리즘 2 번**
  - ■

- 알고리즘 1 번 | Minor2 Reward Only PPO
  - 모델별 적용 결과

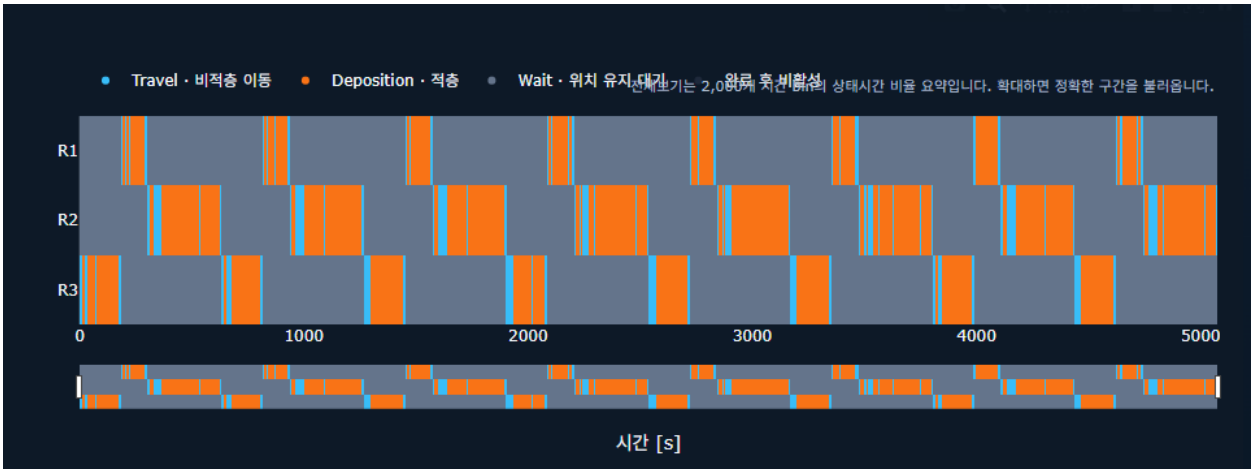
모델명	Makespan (DD:HH:MM:SS)	계산 시간 (DD:HH:MM:SS)	Validator 적용 결과 (Pass or Fail)
01 (Honeycomb)	00:00:40:50	00:00:00:05	PASS
02 (Spiral)	00:01:24:34	00:00:00:10	PASS
03 (Triangular truss)	00:01:06:16	00:00:00:06	PASS
04 (Motor inner)	00:19:16:04	00:00:01:25	PASS
05 (Twisted shell)	04:12:27:22	00:00:07:14	PASS
06 (Angle frame)	00:02:41:07	00:00:00:16	PASS

  - Validator 적용 결과 캡쳐 화면

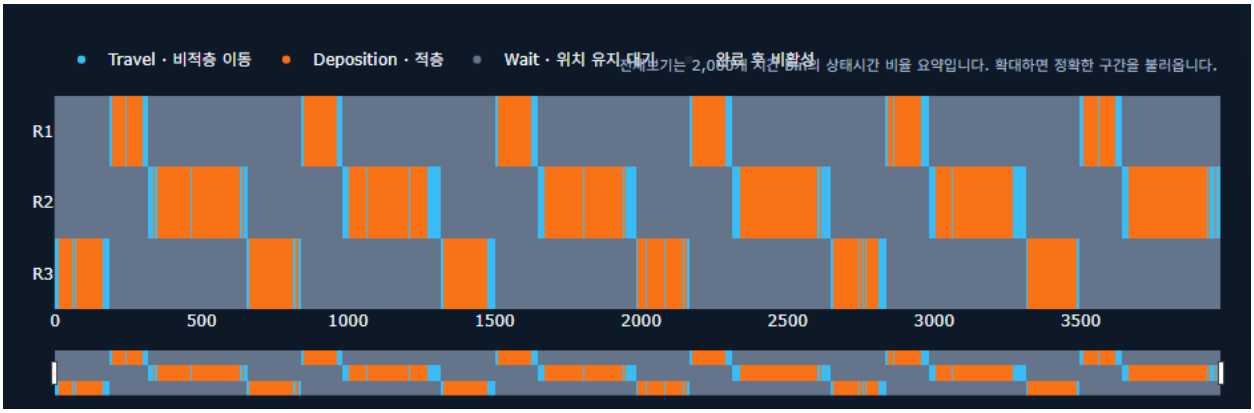
◆ 01



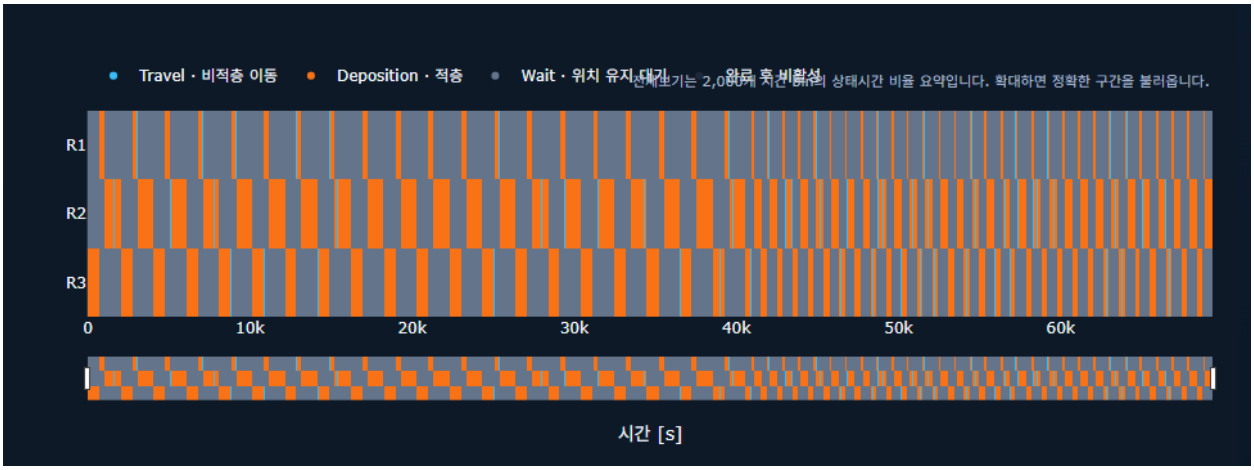
◆ 02



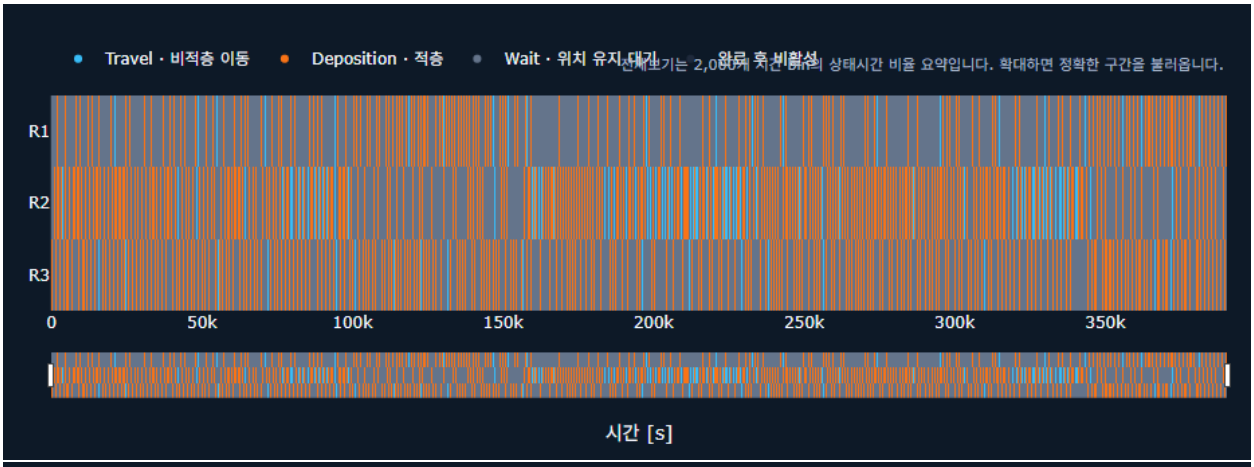
◆ 03

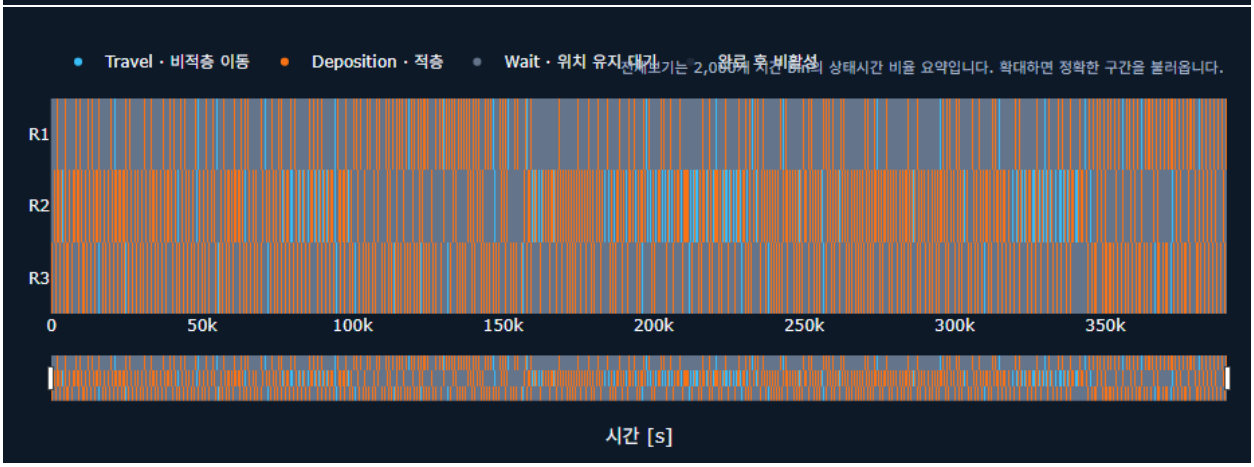

Travel · 비적층 이동
Deposition · 적층
Wait · 위치 유지 대기
전체보기는 2,000개 처리 bin의 상태시간 비율 요약입니다. 확대하면 정확한 구간을 불러옵니다.
R1
R2
R3
0
500
1000
1500
2000
2500
3000
3500

◆ 04

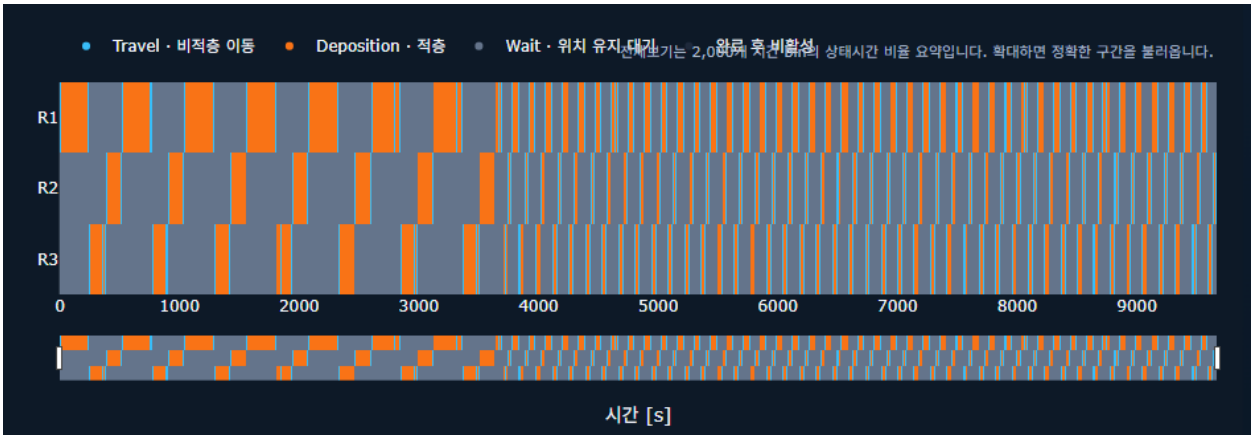

Travel · 비적층 이동
Deposition · 적층
Wait · 위치 유지 대기
전체보기는 2,000개 처리 bin의 상태시간 비율 요약입니다. 확대하면 정확한 구간을 불러옵니다.
R1
R2
R3
0
10k
20k
30k
40k
50k
60k
시간 [s]

◆ 05


Travel · 비적층 이동
Deposition · 적층
Wait · 위치 유지 대기
전체보기는 2,000개 지점 sm의 상태시간 비율 요약입니다. 확대하면 정확한 구간을 불러옵니다.
R1
R2
R3
0
50k
100k
150k
200k
250k
300k
350k
시간 [s]


Travel · 비적층 이동
Deposition · 적층
Wait · 위치 유지 대기
전체보기는 2,000개 지점 sm의 상태시간 비율 요약입니다. 확대하면 정확한 구간을 불러옵니다.
R1
R2
R3
0
50k
100k
150k
200k
250k
300k
350k
시간 [s]

◆ 06


Travel · 비적층 이동
Deposition · 적층
Wait · 위치 유지 대기
전체보기는 2,000개 지점 sm의 상태시간 비율 요약입니다. 확대하면 정확한 구간을 불러옵니다.
R1
R2
R3
0
1000
2000
3000
4000
5000
6000
7000
8000
9000
시간 [s]

● 수도코드

■

Minor2 - reward-only layer PPO

Paper-style algorithm | three-robot path generation and IoU-gated time/utilization reward

Algorithm 2. Minor2 - reward-only layer PPO

```
Input: training STL/config set D_tr; held-out validation set D_val; initial
      Main PPO weights
Initialize layer-path policy pi_theta; retain original action and
      route-generation environment
For PPO update k = 1, ..., K do
  Collect one rollout of n_steps layer transitions from training STL
      episodes
  For each layer, observe geometry/raster, schedule progress, robot reach
      and process settings
    Select hatch angle, spacing, start corner, reversal, partition axis,
      quota and robot order
    Generate contours/hatch and partition segments among reachable robots
    Build concurrent routes; check reach/collision; retry serialized starts
      on collision
    Reject infeasible paths with invalid-action penalty -2
    Otherwise compute deposited geometry, layer IoU I, duration T and
      utilization U_D
    If I > 0.9, reward = 10 * U_D * reference_time / T
    Else terminal Validator PASS = +1; FAIL/ERROR = -1; no result = 0
    Append layer schedule; return next observation, reward and completion
      flag
  End for; update pi_theta with PPO
  Every four updates evaluate held-out STLs; select checkpoint by Validator
      pass rate
End for; return best path policy and export training-STL trajectories
```

- ㄴ이 알고리즘을 이 문제에 적용한 이유
  - ◆
- 알고리즘에 대한 세부 설명(특징, 장단점 등)
  - ◆

■ 기타

■ Parameters

파라미터	현재 설정	의미
학습 업데이트	40 / 40 회	요청 횟수 / 완료 횟수
학습 타임스텝	2560 / 2560	완료 / 요청
n_steps	64	PPO rollout 단계 수
batch_size	32	미니배치 크기
learning_rate	0.0003	학습률
gamma	0.99	할인율
gae_lambda	0.95	GAE 람다
seed	42	재현용 시드
device	cuda	정책 네트워크 계산 장치
early_stopping	-	조기 종료 설정
eval_every_updates	4	평가 주기
eval_episodes	1	평가 STL 별 반복 횟수
geometry_device	cpu	기하·Validator 계산 장치
training STL	6	학습 입력 STL 개수
validation STL	6	학습 중 평가 STL 개수

placement_enabled	False	전역 XY placement PPO 사용 여부
reward_formula	iou_gt_0.9_deposition_utilization_over_layer_makespan_v1	학습 보상식 ID
best_step	256	최고 검증 점수 체크포인트 단계
best_model	best_model.zip	실행 시 불러오는 모델
training_status	completed	학습 기록 상태



- **알고리즘 2 번**
  - ■

Trajectory files appendix

Trajectory CSV files are stored by experiment and input STL. The experiment folders retain their original evaluation trees; separate 1.0.2 exports are listed below.

Main 1.0.2

main/stl_files/job_00-job_05/episode_000/input/trajectory.csv; main/validation_data/job_00-job_05/episode_000/input/trajectory.csv

Minor1

minor1/stl_files/job_00-job_05/episode_000/input/trajectory.csv; minor1/validation_data/job_00-job_05/episode_000/input/trajectory.csv

Minor2

minor2/stl_files/job_00-job_05/episode_000/input/trajectory.csv; minor2/validation_data/job_00-job_05/episode_000/input/trajectory.csv

Main standalone exports

main/stl_files/exported_trajectory/01-06/trajectory.csv

Main 1.0.2 - preprocessing and PPO training

Paper-style algorithm | STL preprocessing, layer actions, idle-time reward and final Validator

Algorithm 1. Main 1.0.2 - preprocessing and PPO training

```
Input: training STL/config set D_tr; held-out validation set D_val; PPO
      rollout settings
For each job, parse config, load/repair STL mesh, validate geometry, process
      and robot limits
Select evaluation layers; slice each section at build_plane + (k + 0.5) *
      layer_height
Normalize layer polygons, discard empty layers, and cache a 16 x 16
      fractional-area raster
Initialize three robots at home and PPO policy pi_theta with
      MultiDiscrete([8,3,4,2,2,3,6])
For PPO update u = 1, ..., U do
  Reset a training STL episode; initialize schedule and cumulative deposited
      geometry
  For each target layer k do
    Observe progress, makespan, IoU, layer geometry/raster, robot reach and
      process settings
    Select hatch angle, spacing, start corner, reversal, partition axis,
      quota and robot order
    Generate inset contours and clipped hatch; split long segments and
      assign reachable robots
    Build travel/deposition/wait routes; check reach, trajectory rules and
      collisions
    On collision retry serialized starts; reject infeasible routes with
      reward -2
    Compute deposited geometry, layer IoU, duration and robot utilization
    Set idle_factor = 1 + sum(robot idle seconds) / layer duration
    Reward = IoU - 0.1 * duration * idle_factor / reference_time
    On final layer run whole-schedule Validator; add +1 for PASS or -1 for
      FAIL/ERROR
    Append routes with time offsets; update observation and PPO rollout
      buffer
  End for; PPO-update policy and periodically score held-out validation
      checkpoints
End for; export best policy, trajectories, validation JSON and Gantt charts
```

Notation and implementation notes

- Preprocessing loads, repairs, validates, slices and normalizes each target STL before training.
- One PPO step generates one complete layer path; three robots share the layer schedule.
- The official whole-trajectory WAAM Validator is the final authority.

Minor1 - joint placement and layer PPO

Paper-style algorithm | shared STL translation, learned path policy and collision-aware schedule search

Algorithm 1. Minor1 - joint placement and layer PPO

```
Input: disjoint training jobs D_tr; held-out validation set D_val; layer
      policy pi_l; placement policy pi_p
Initialize pi_l and pi_p (optionally load compatible weights); initialize
      layer and placement environments
For PPO update u = 1, ..., U do
  Collect n_steps layer transitions under current placement; PPO-update
      layer/path policy pi_l
  If u is a placement-update interval, collect n_place full-STL episodes
      (n_place >= 2)
  For each placement episode, select training STL q and observe
      mesh/workspace/robot features
    Sample shared (dx,dy) from pi_p; clip and scale by workspace radius
    Project translation so every mesh-hull vertex remains inside the
      workspace
    Translate the entire STL by (dx,dy,0) before slicing; all layers share
      the shift
    For each target layer k, pi_l selects path parameters and robot
      partition; generate segments
      Build original concurrent schedule; use serial fallback if collision
      requires it
      Try nearest-base/load-balanced assignment and reorder routes
      near-to-far
      Try concurrent starts and bounded staggered offsets in two robot
      priority orders
      Reject candidates failing reach, trajectory or collision checks;
      retain shortest feasible
      Keep all generated deposition segments; search changes
      assignment/order/timing only
      If IoU > 0.9, r = 10 * mean deposition utilization * reference time /
      layer duration
      Else terminal Validator PASS = +1; FAIL/ERROR = -1; unavailable = 0
    End for; placement reward = mean layer rewards from this STL only
  End for; PPO-update pi_p from full-STL placement rewards
  Evaluate held-out jobs; retain best paired checkpoint by validation key
End for; export full trajectories and run official WAAM Validator
```

Notation and implementation notes

- Two PPO policies are learned: global XY placement and layer/path selection.
- One shared XY shift is applied to the complete STL before slicing.
- The collision-aware schedule candidate search is bounded heuristic search, not a learned scheduler.

Minor2 - reward-only layer PPO

Paper-style algorithm | three-robot path generation and IoU-gated time/utilization reward

Algorithm 2. Minor2 - reward-only layer PPO

```
Input: training STL/config set D_tr; held-out validation set D_val; initial
      Main PPO weights
Initialize layer-path policy pi_theta; retain original action and
      route-generation environment
For PPO update k = 1, ..., K do
  Collect one rollout of n_steps layer transitions from training STL
      episodes
  For each layer, observe geometry/raster, schedule progress, robot reach
      and process settings
    Select hatch angle, spacing, start corner, reversal, partition axis,
      quota and robot order
    Generate contours/hatch and partition segments among reachable robots
    Build concurrent routes; check reach/collision; retry serialized starts
      on collision
    Reject infeasible paths with invalid-action penalty -2
    Otherwise compute deposited geometry, layer IoU I, duration T and
      utilization U_D
    If I > 0.9, reward = 10 * U_D * reference_time / T
    Else terminal Validator PASS = +1; FAIL/ERROR = -1; no result = 0
    Append layer schedule; return next observation, reward and completion
      flag
  End for; update pi_theta with PPO
  Every four updates evaluate held-out STLs; select checkpoint by Validator
      pass rate
End for; return best path policy and export training-STL trajectories
```

Notation and implementation notes

- Minor2 changes layer rewards and has no global STL-placement policy.
- The performance reward applies only when layer IoU is strictly greater than 0.9.
- Mean deposition utilization includes deposition time only; travel and waiting are excluded.
- The whole-STL Validator confirms final schedule validity.

Minor1 과 1.1.1.e retry1 결과 비교

동일한 원본 학습 STL 6 개를 Minor1 최우수 체크포인트(1,024 step)와 1.1.1.e retry1 최종 체크포인트(2,560 step)로 평가한 결과이다. 두 모델의 정책, STL 배치 및 학습량이 다르므로 표의 차이는 방법 전체의 결과 차이다.

학습 STL Makespan

STL	**Minor1** **DD:HH:MM:SS**	**Retry1** **DD:HH:MM:SS**	**변화율**	**Validator** **Minor1 / Retry1**
01 Honeycomb	00:00:35:03	00:00:44:08	+25.90%	PASS / PASS
02 Spiral	00:01:09:18	00:01:39:54	+44.13%	PASS / PASS
03 Triangular truss	00:00:57:30	00:01:09:48	+21.38%	PASS / PASS
04 Motor inner	00:15:02:29	00:17:52:37	+18.85%	PASS / PASS
05 Twisted shell	02:03:13:53	01:14:20:19	-25.17%	PASS / PASS
06 Angle frame	00:02:12:28	00:02:26:40	+10.73%	PASS / PASS

변화율 = (Retry1 Makespan / Minor1 Makespan − 1) × 100. 음수는 Retry1 의 제작시간 감소를 뜻한다.

사례별 계산 시간

STL	**Minor1** **DD:HH:MM:SS**	**Retry1** **DD:HH:MM:SS**
01	00:00:00:13	00:00:00:10
02	00:00:00:25	00:00:00:43
03	00:00:00:14	00:00:00:17
04	00:00:03:00	00:00:04:05
05	00:00:37:07	00:00:04:56
06	00:00:01:27	00:00:00:10

기록된 계산시간의 측정 구간은 버전별 실행기에 따라 다르다. Retry1 의 값은 trajectory 생성 구간이며 WAAM Validator 실행시간을 제외한다. 정확한 초 단위 값은 minor1_retry1_comparison.csv 에 있다.

공통 validation 6 개 Makespan

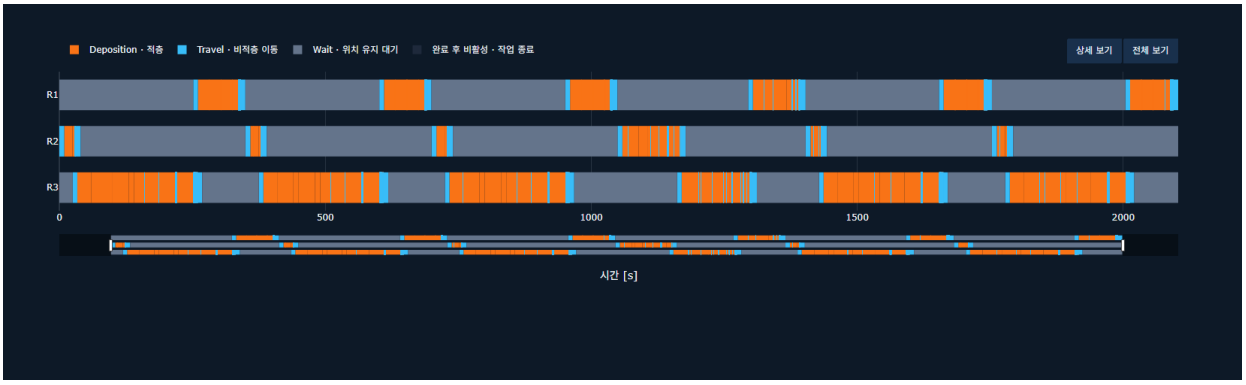
입력	Minor1 DD:HH:MM:SS	Retry1 DD:HH:MM:SS	변화율	Validator
00_sample	00:00:00:43	00:00:01:02	+43.95%	PASS / PASS
01_rectangle	00:00:08:20	00:00:09:13	+10.42%	PASS / PASS
02_triangle	00:00:03:02	00:00:03:21	+10.44%	PASS / PASS
03_l_shape	00:00:08:03	00:00:08:54	+10.73%	PASS / PASS
04_cross	00:00:04:17	00:00:04:29	+4.79%	PASS / PASS
05_ring	00:00:05:10	00:00:05:57	+15.11%	PASS / PASS

작성자 설명

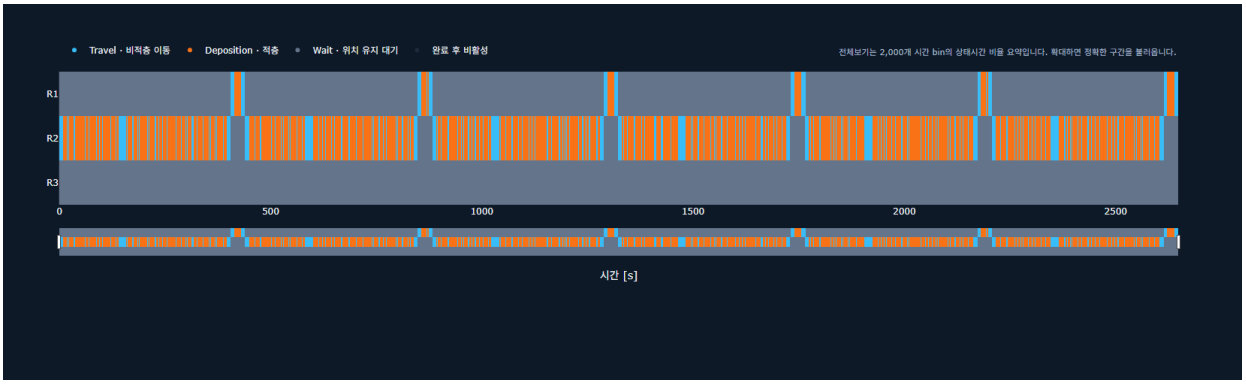
Minor1 과 Retry1 WAAM Validator 간트 비교

01 Honeycomb | Minor1 00:00:35:03 | Retry1 00:00:44:08

Minor1



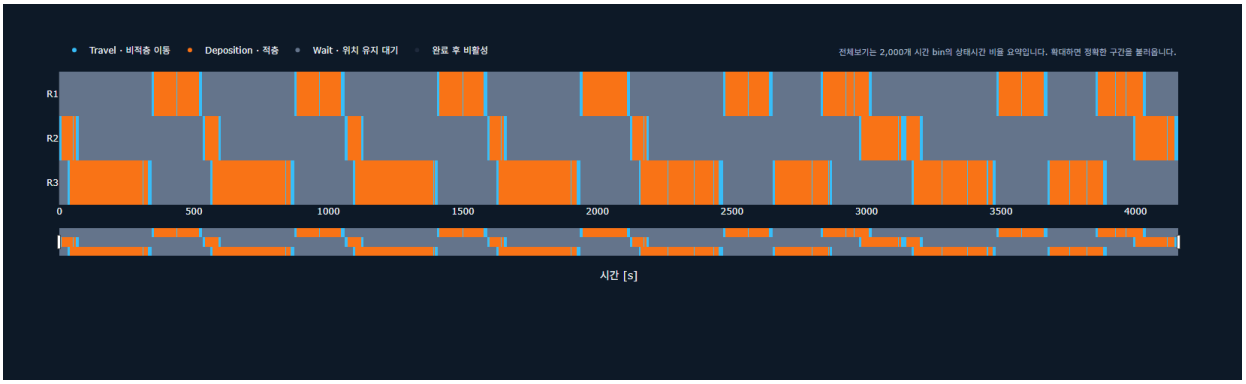
1.1.1.e retry1



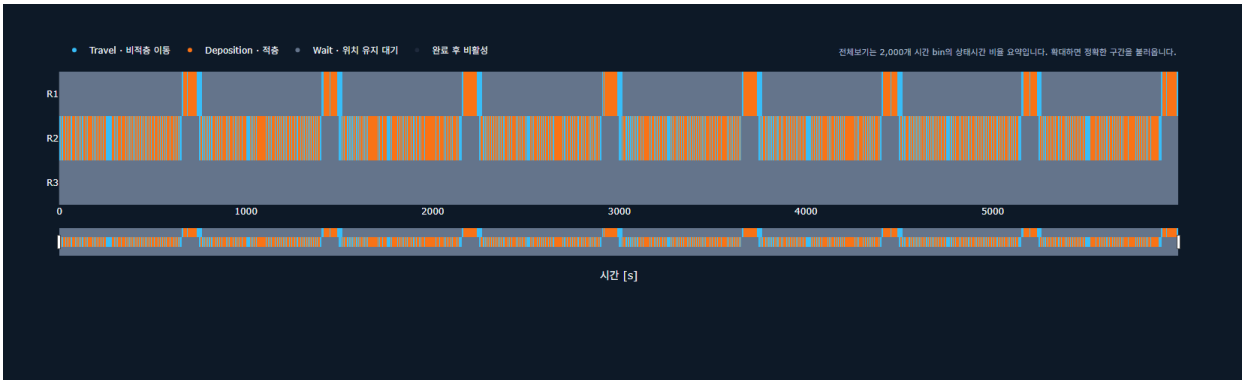
Minor1 과 Retry1 WAAM Validator 간트 비교

02 Spiral | Minor1 00:01:09:18 | Retry1 00:01:39:54

Minor1



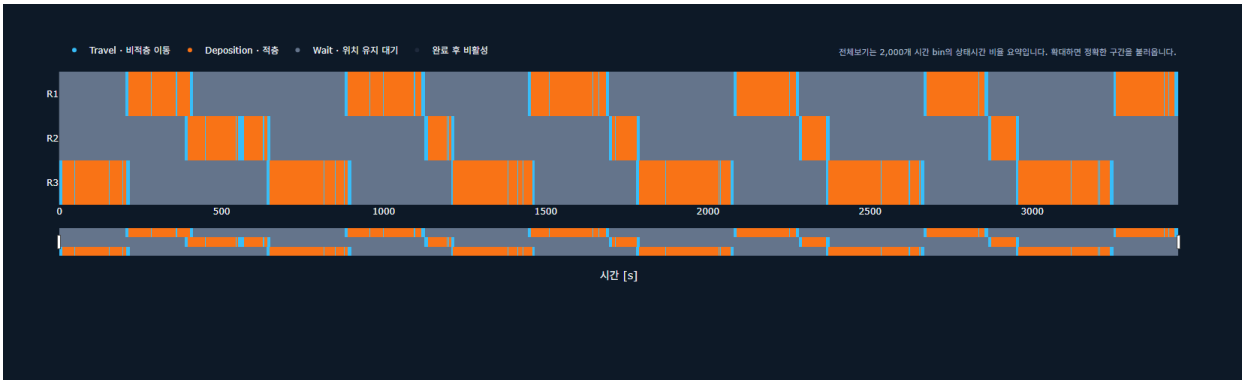
1.1.1.e retry1



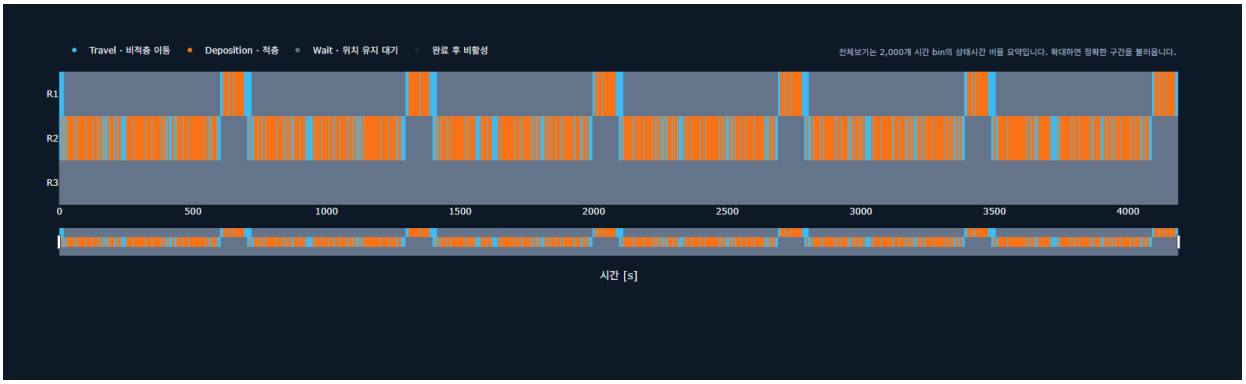
Minor1 과 Retry1 WAAM Validator 간트 비교

03 Triangular truss | Minor1 00:00:57:30 | Retry1 00:01:09:48

Minor1



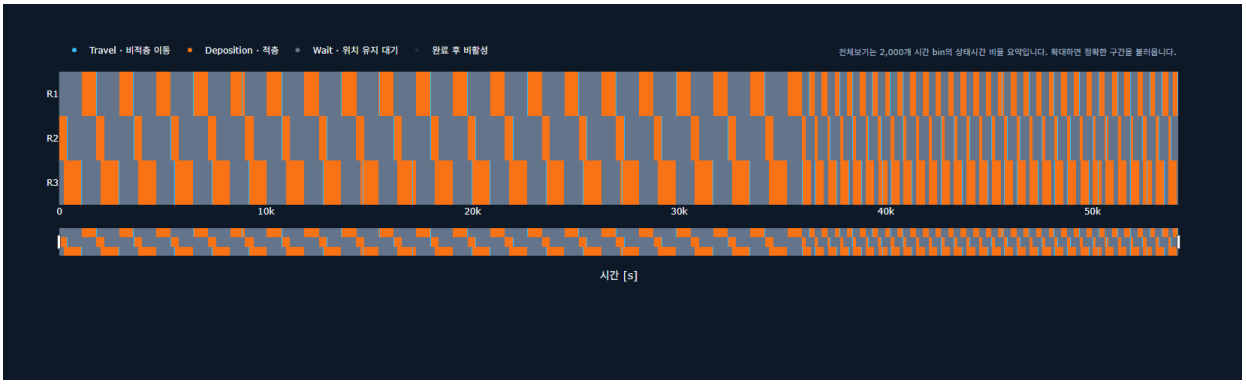
1.1.1.e retry1



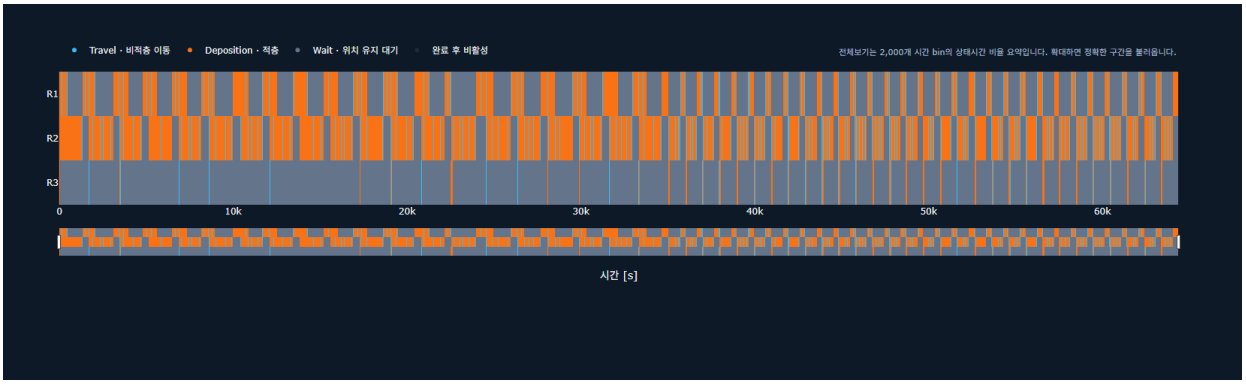
Minor1 과 Retry1 WAAM Validator 간트 비교

04 Motor inner | Minor1 00:15:02:29 | Retry1 00:17:52:37

Minor1



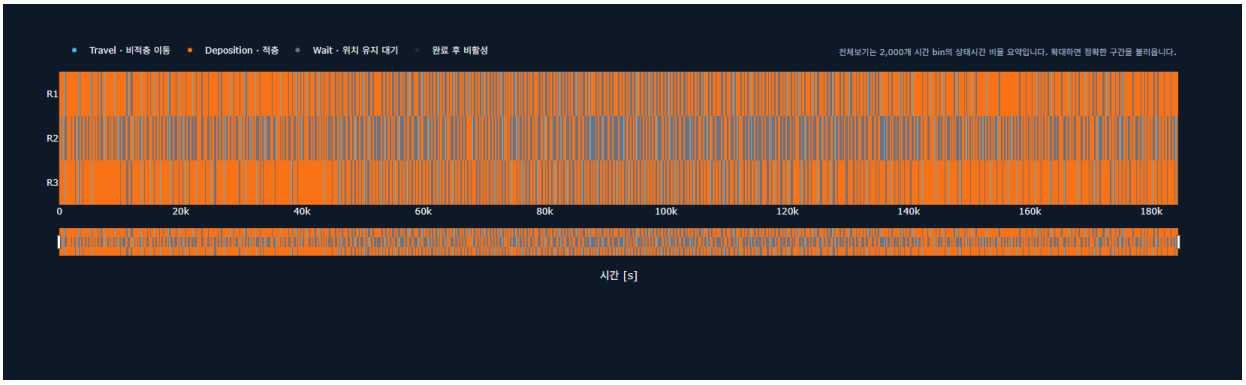
1.1.1.e retry1



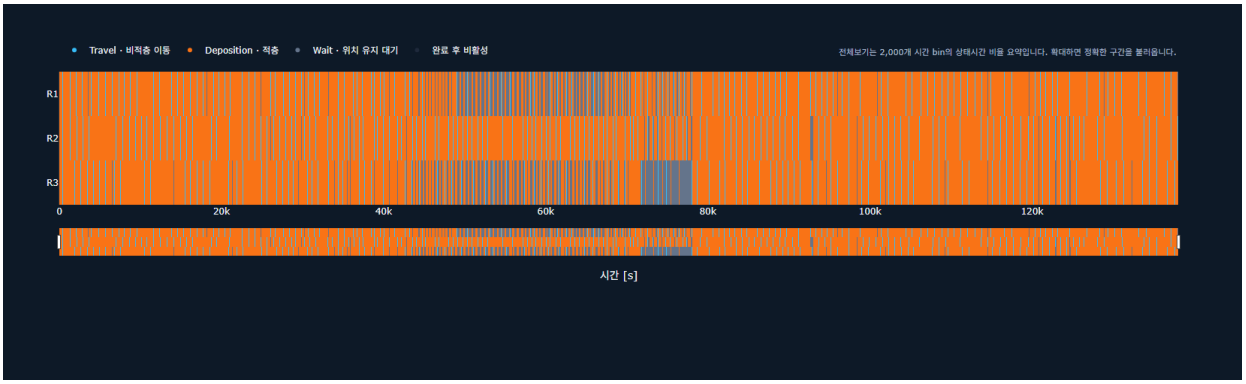
Minor1 과 Retry1 WAAM Validator 간트 비교

05 Twisted shell | Minor1 02:03:13:53 | Retry1 01:14:20:19

Minor1



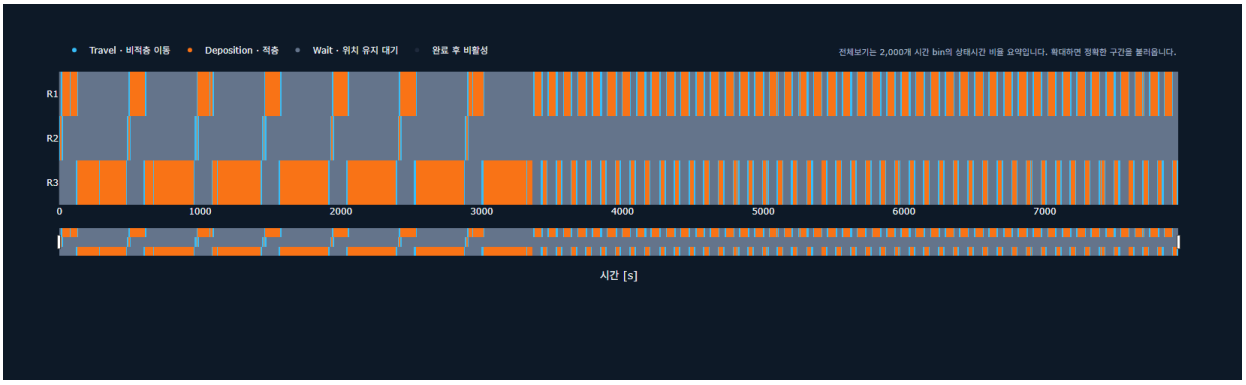
1.1.1.e retry1



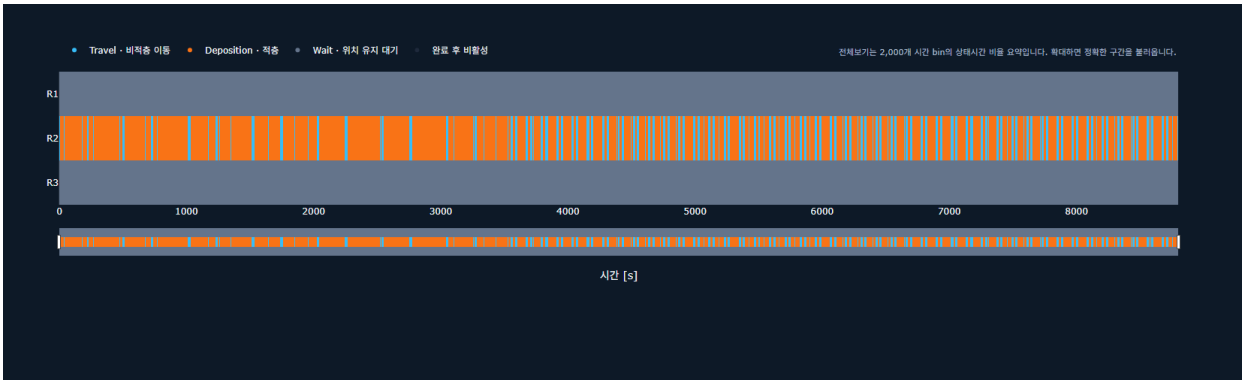
Minor1 과 Retry1 WAAM Validator 간트 비교

06 Angle frame | Minor1 00:02:12:28 | Retry1 00:02:26:40

Minor1



1.1.1.e retry1



Algorithm 1

Projected placement and retry-before-handoff layer scheduling

Input: STL M; robot configuration C; train set Dtr; validation set Dval

Output: trained PPO policy π and collision-checked trajectories

Preprocessing and layer environment

Sample interior solid points from M; assign weight 1 to every 3D sample.

Project samples to XY and retain multiplicity from different Z heights.

Search feasible rigid XY offsets; reject placements outside workspace or reach union.

For each feasible offset, assign each projected point to its nearest reachable robot.

Choose the offset minimizing the standard deviation of the three point shares; break ties by maximum share error, travel cost, then offset magnitude.

Apply the chosen single XY translation to the whole STL; slice the placed mesh.

For each layer, generate fixed-spacing hatch segments at the selected hatch angle.

Split segments at reach-circle boundaries and nearest-base bisectors; assign every resulting segment to its nearest reachable robot.

PPO action and path construction

Policy selects hatch angle (8), each robot start XY (33 × 33 grid), each sweep direction (CW/CCW), and one of 6 layer-priority permutations.

Project each requested start onto that robot's owned segment; split there, then order its segments by the selected angular sweep.

Collision retry and scheduling

Visit robots in priority order. For each owned task, test approach, deposition, and provisional safe return against other robots' reserved trajectories.

If safe, reserve the task. If blocked, keep it pending and test other owned tasks.

Repeat a pending sweep after any task is accepted; stop when none remain blocked or a sweep accepts no task.

Transfer only unresolved tasks to the highest-priority reachable robot; if none qualifies, retain them in the original robot's deferred queue.

Append deferred work after primary routes; delay its release from home only when required for collision clearance. Preserve every task and deposition length.

Check task coverage, physical collision clearance, and trajectory semantics.

Training and evaluation

Set layer reward $r_\ell = -T_\ell$; episode return = $-\sum_\ell T_\ell$ = −makespan.

Update PPO for 40 updates × 64 steps; evaluate periodically and retain the checkpoint with the lowest validation mean makespan. No early stopping.

Run the official WAAM Validator for evaluation diagnostics only; it is not reward.

Reward: $r_\ell = -T_\ell$; episode return = −makespan | Validator: evaluation only